



UNIVERSITI MALAYSIA TERENGGANU

FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONORS

SEMESTER II 2025/2026

CSA3023

WEB-BASED APPLICATION DEVELOPMENT

LAB 4:

JSP: SCRIPTLET, EXPRESSION & STANDARD ACTIONS

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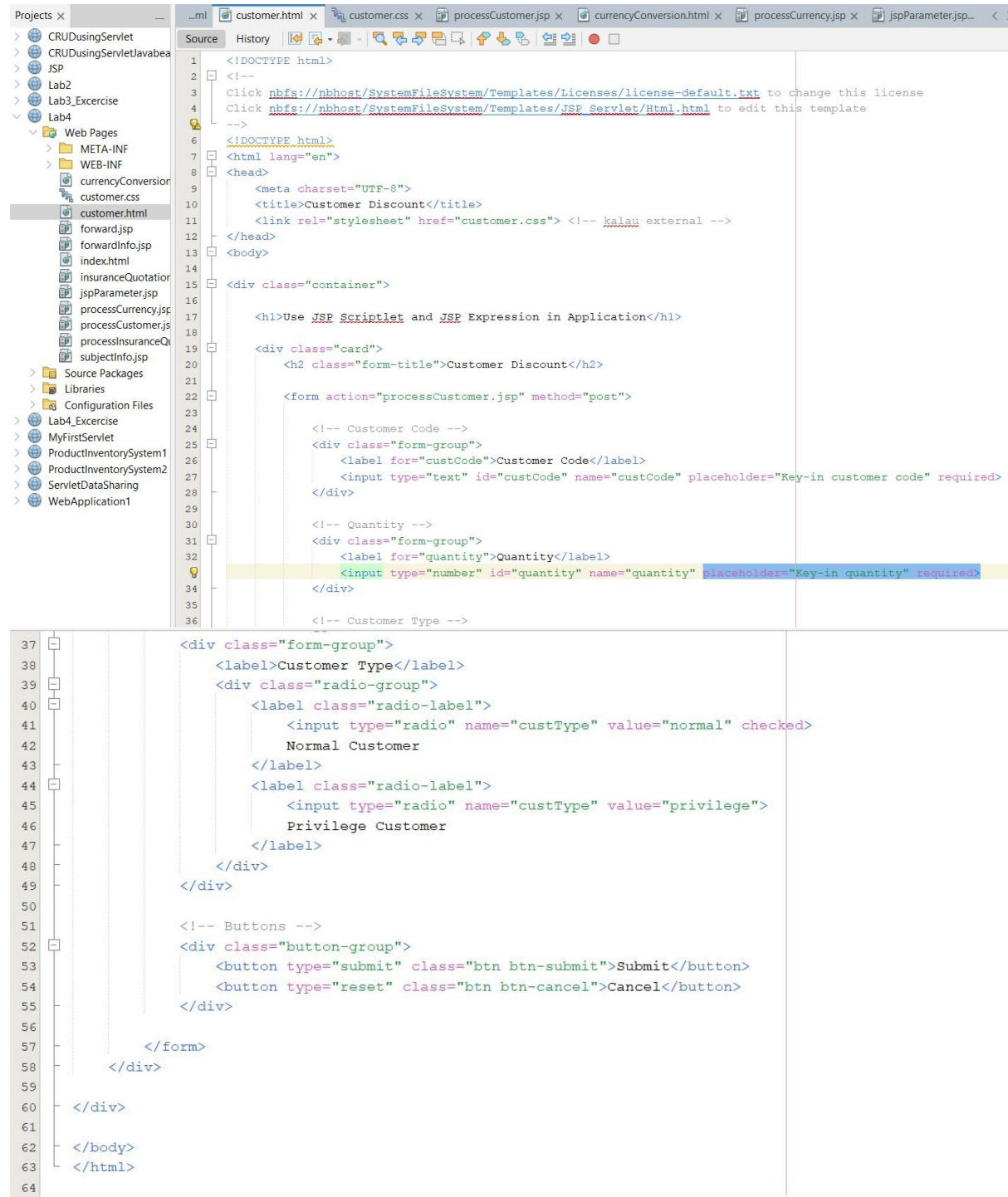
Date:

21 APRIL 2026

Task 1: Using JSP Scripting

Objective: Apply JSP Scriptlet and JSP Expression to process input data and display dynamic output.

Answer:



```
1 <!DOCTYPE html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Html.html to edit this template
5 -->
6 <!DOCTYPE html>
7 <html lang="en">
8 <head>
9 <meta charset="UTF-8">
10 <title>Customer Discount</title>
11 <link rel="stylesheet" href="customer.css"> <!-- kalau external -->
12 </head>
13 <body>
14
15 <div class="container">
16
17 <h1>Use JSP Scriptlet and JSP Expression in Application</h1>
18
19 <div class="card">
20 <h2 class="form-title">Customer Discount</h2>
21
22 <form action="processCustomer.jsp" method="post">
23
24 <!-- Customer Code -->
25 <div class="form-group">
26 <label for="custCode">Customer Code</label>
27 <input type="text" id="custCode" name="custCode" placeholder="Key-in customer code" required>
28 </div>
29
30 <!-- Quantity -->
31 <div class="form-group">
32 <label for="quantity">Quantity</label>
33 <input type="number" id="quantity" name="quantity" placeholder="Key-in quantity" required>
34 </div>
35
36 <!-- Customer Type -->
37 <div class="form-group">
38 <label>Customer Type</label>
39 <div class="radio-group">
40 <label class="radio-label">
41 <input type="radio" name="custType" value="normal" checked>
42 Normal Customer
43 </label>
44 <label class="radio-label">
45 <input type="radio" name="custType" value="privilege">
46 Privilege Customer
47 </label>
48 </div>
49 </div>
50
51 <!-- Buttons -->
52 <div class="button-group">
53 <button type="submit" class="btn btn-submit">Submit</button>
54 <button type="reset" class="btn btn-cancel">Cancel</button>
55 </div>
56
57 </form>
58 </div>
59
60 </div>
61
62 </body>
63 </html>
64
```

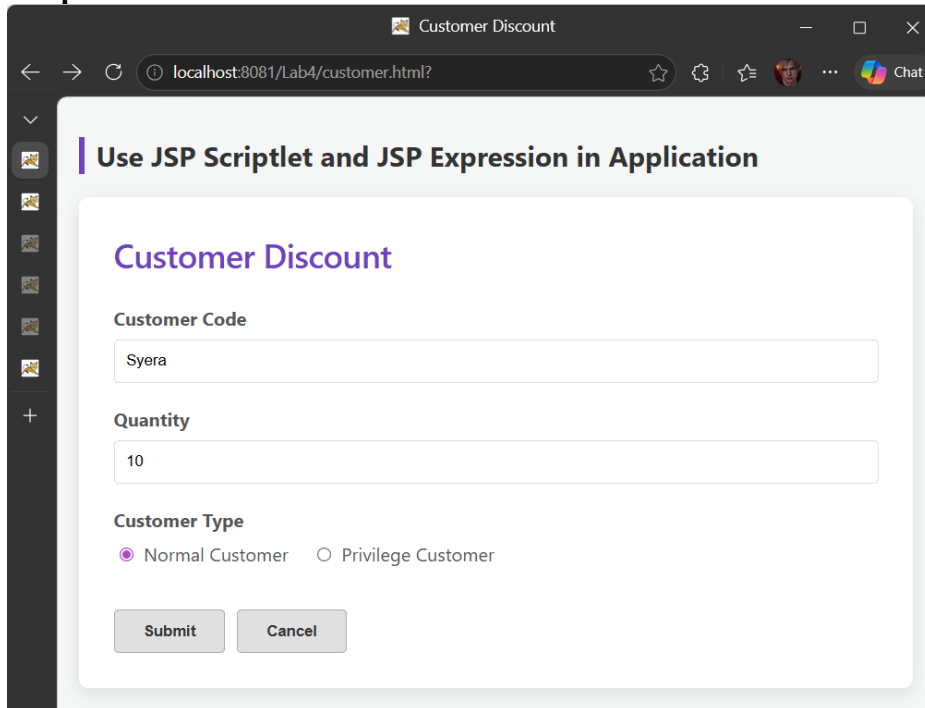
```
...ml customer.html x customer.css x processCustomer.jsp x currencyConversion.html x processCurrency.jsp x jspParameter.j
Source History
1 <%--
2     Document    : processCustomer
3     Created on  : 21 Apr 2026, 2:50:33 pm
4     Author     : syaheera
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11     <title>Customer Discount Result</title>
12     <link rel="stylesheet" href="customer.css">
13 </head>
14 <body>
15
16 <div class="container">
17
18     <h1>Customer Discount Result</h1>
19
20     <div class="card">
21         <h2 class="form-title">Transaction Summary</h2>
22
23         <%
24             final double price = 10.0;
25
26             String cust_no = request.getParameter("custCode");
27             String cust_type = request.getParameter("custType");
28
29             int quantity = 0;
30             try {
31                 quantity = Integer.parseInt(request.getParameter("quantity"));
32             } catch (Exception e) {
33                 quantity = 0;
34             }
35
36             double total = 0;
```

```

37     String message = "";
38
39     if ("normal".equals(cust_type) && quantity > 100) {
40         message = "You're entitled to 10% discount";
41         total = quantity * price * 0.9;
42     }
43     else if ("privilege".equals(cust_type) && quantity > 100) {
44         message = "You're entitled to 25% discount";
45         total = quantity * price * 0.75;
46     }
47     else {
48         message = "You're not entitled to any discount";
49         total = quantity * price;
50     }
51
52     String custTypeDisplay = "normal".equals(cust_type) ?
53         "Normal Customer" : "Privilege Customer";
54     %>
55
56     <!-- OUTPUT STYLE MACAM GAMBAR -->
57     <div class="result-group">
58         <label>Customer Code:</label>
59         <p><%= cust_no %></p>
60     </div>
61
62     <div class="result-group">
63         <label>Quantity:</label>
64         <p><%= quantity %></p>
65     </div>
66
67     <div class="result-group">
68         <label>Customer Type:</label>
69         <p><%= custTypeDisplay %></p>
70     </div>
71
72     <div class="result-group">
73         <label>Status:</label>
74         <p><%= message %></p>
75     </div>
76
77     <div class="result-group">
78         <label>Total Amount:</label>
79         <p>RM <%= String.format("%.2f", total) %></p>
80     </div>
81
82     <br>
83
84     <form action="customer.html">
85         <button class="btn btn-back">Back</button>
86     </form>
87
88     </div>
89
90 </div>
91
92 </body>
93 </html>

```

Output:



The screenshot shows a web browser window titled "Customer Discount". The address bar displays "localhost:8081/Lab4/customer.html?". The page content includes a sidebar with icons, a main heading "Use JSP Scriptlet and JSP Expression in Application", and a form titled "Customer Discount". The form contains three input fields: "Customer Code" with the value "Syera", "Quantity" with the value "10", and "Customer Type" with radio buttons for "Normal Customer" (selected) and "Privilege Customer". At the bottom of the form are "Submit" and "Cancel" buttons.

Customer Discount

Customer Code

Syera

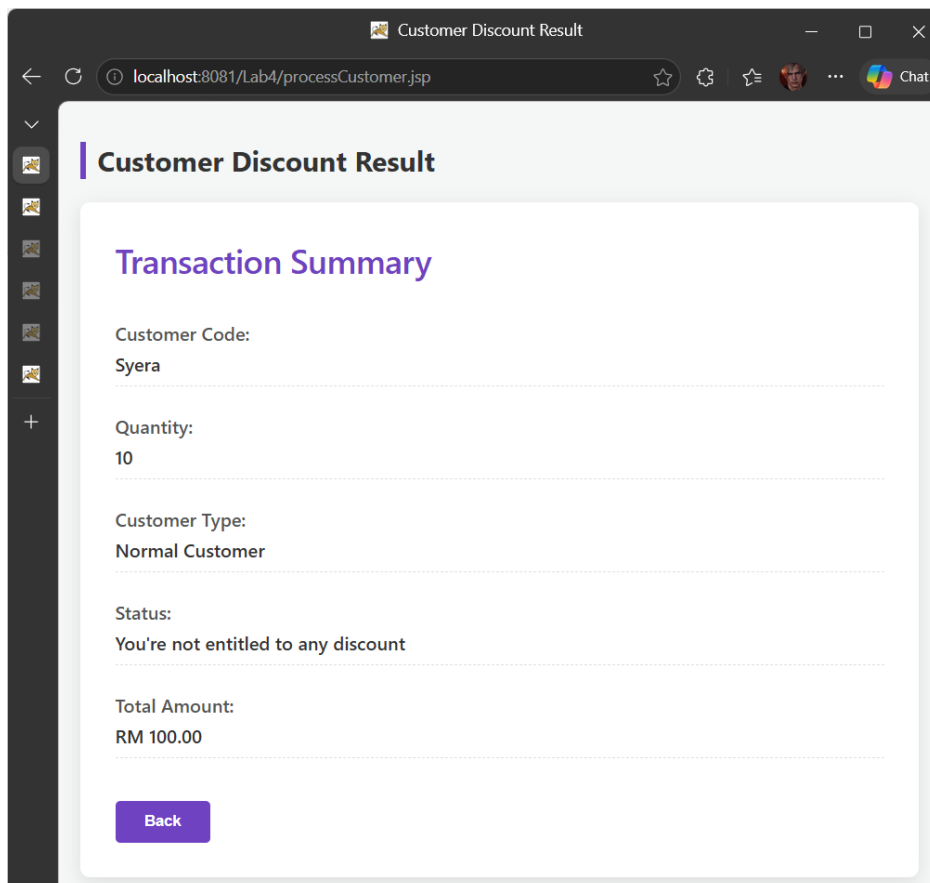
Quantity

10

Customer Type

☒ Normal Customer ☐ Privilege Customer

Submit Cancel



The screenshot shows a web browser window titled "Customer Discount Result". The address bar displays "localhost:8081/Lab4/processCustomer.jsp". The page content includes a sidebar with icons, a main heading "Customer Discount Result", and a form titled "Transaction Summary". The form displays the input data: "Customer Code: Syera", "Quantity: 10", "Customer Type: Normal Customer", and "Status: You're not entitled to any discount". At the bottom, it shows "Total Amount: RM 100.00" and a "Back" button.

Customer Discount Result

Transaction Summary

Customer Code:

Syera

Quantity:

10

Customer Type:

Normal Customer

Status:

You're not entitled to any discount

Total Amount:

RM 100.00

Back

Reflection

1. What have you learnt from this exercise?

Answer:

i have learned how to use JSP Scriptlets to retrieve and process form data on the server side using `request.getParameter()` and how to display that processed dynamic content back to the user using JSP Expressions.

2. Explain three (3) types of JSP scripting elements.

Answer:

- Scriptlets (`<% ... %>`): Used to embed Java code directly into a JSP page to implement core programming and business logic.
- Expressions (`<%= ... %>`): Used to convert a Java expression into a string and insert it directly into the output HTML sent to the browser.
- Declarations (`<%! ... %>`): Used to define variables, constants, or reusable methods that can be accessed by other parts of the JSP page.

Task 2: Using JSP (Scripting, Declaration and Expression)

Objective: Apply JSP Declaration, JSP Scriptlet, and JSP Expression to perform currency conversion in a web application

Answer:

```
...ml customer.html x customer.css x processCustomer.jsp x currencyConversion.html x processCurrency.jsp x jspParameter.jsp
Source History
1 <!DOCTYPE html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
5 -->
6 <!DOCTYPE html>
7 <html>
8 <head>
9 <title>Currency Conversion</title>
10 <link rel="stylesheet" href="customer.css">
11 </head>
12 <body>
13
14 <div class="container">
15
16 <h1>Use JSP Declaration Tag, JSP Scriptlet and JSP Expression in Application</h1>
17
18 <div class="card">
19 <h2 class="form-title">Currency Conversion</h2>
20
21 <form action="processCurrency.jsp" method="post">
22
23 <!-- Amount -->
24 <div class="form-group">
25 <label>Amount (in RM)</label>
26 <input type="number" name="amount" placeholder="In RM" required>
27 </div>
28
29 <!-- Currency -->
30 <div class="form-group">
31 <label>Convert to</label>
32 <select name="currency">
33 <option value="1">USD</option>
34 <option value="2">EURO</option>
35 <option value="3">JPY</option>
36 <option value="4">SGD</option>
37 </select>
38 </div>
39
40 <!-- Button -->
41 <div class="button-group">
42 <button type="submit" class="btn btn-submit">Submit</button>
43 <button type="reset" class="btn btn-cancel">Cancel</button>
44 </div>
45
46 </form>
47 </div>
48
49 </div>
50
51 </body>
52 </html>
53
```

```
...ml customer.html x customer.css x processCustomer.jsp x currencyConversion.html x processCurrency.jsp x
Source History
1 <%--
2     Document    : processCurrency
3     Created on  : 21 Apr 2026, 3:22:41 pm
4     Author     : syaheera
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <%!
9     final double USD = 0.25;
10    final double EURO = 0.21;
11    final double JPY = 40;
12    final double SGD = 0.32;
13
14    private double calculateRate(String currency, double amount) {
15        if (currency == null) return 0;
16
17        switch(currency) {
18            case "1": return amount * USD;
19            case "2": return amount * EURO;
20            case "3": return amount * JPY;
21            case "4": return amount * SGD;
22        }
23        return 0;
24    }
25 %>
26
27 <!DOCTYPE html>
28 <html>
29     <head>
30         <title>Currency Result</title>
31         <link rel="stylesheet" href="customer.css">
32     </head>
33     <body>
34
35         <div class="container">
36
```



```

37     <h1>Currency Conversion Result</h1>
38
39     <div class="card">
40
41         <%
42             String currencyType = request.getParameter("currency");
43             String amountRaw = request.getParameter("amount");
44
45             double amount = 0;
46             double total = 0;
47
48             try {
49                 if (amountRaw != null) {
50                     amount = Double.parseDouble(amountRaw);
51                     total = calculateRate(currencyType, amount);
52                 }
53             } catch (Exception e) {
54                 amount = 0;
55             }
56
57             String currencyName = "N/A";
58             if ("1".equals(currencyType)) currencyName = "USD";
59             else if ("2".equals(currencyType)) currencyName = "EURO";
60             else if ("3".equals(currencyType)) currencyName = "JPY";
61             else if ("4".equals(currencyType)) currencyName = "SGD";
62             %>
63
64         <!-- OUTPUT STYLE -->
65         <div class="result-group">
66             <label>Amount (RM):</label>
67             <p>RM <%= amount %></p>
68         </div>
69
70         <div class="result-group">
71             <label>Converted To:</label>
72             <p><%= currencyName %></p>
73         </div>
74
75         <div class="result-group">
76             <label>Converted Amount:</label>
77             <p><%= String.format("%.2f", total) %></p>
78         </div>
79
80         <br>
81
82         <form action="currencyConversion.html">
83             <button class="btn btn-back">Back</button>
84         </form>
85     </div>
86 </div>
87
88 </body>
89 </html>
90

```

Output:

Currency Conversion

localhost:8081/Lab4/currencyConversion.html

Use JSP Declaration Tag, JSP Scriptlet and JSP Expression in Application

Currency Conversion

Amount (in RM)

Convert to

USD

USD
EURO
JPY
SGD

Currency Result

localhost:8081/Lab4/processCurrency.jsp

Currency Conversion Result

Amount (RM):
RM 100.0

Converted To:
USD

Converted Amount:
25.00

Back

Reflection

1. What have you learnt from this exercise?

Answer:

This exercise demonstrated how to create reusable logic within a web page by using a JSP Declaration tag to define constants and methods (such as a conversion method) and then calling those methods within a scriptlet to perform calculations based on user input.

Task 3: Using JSP Standard Action (Include and Param)

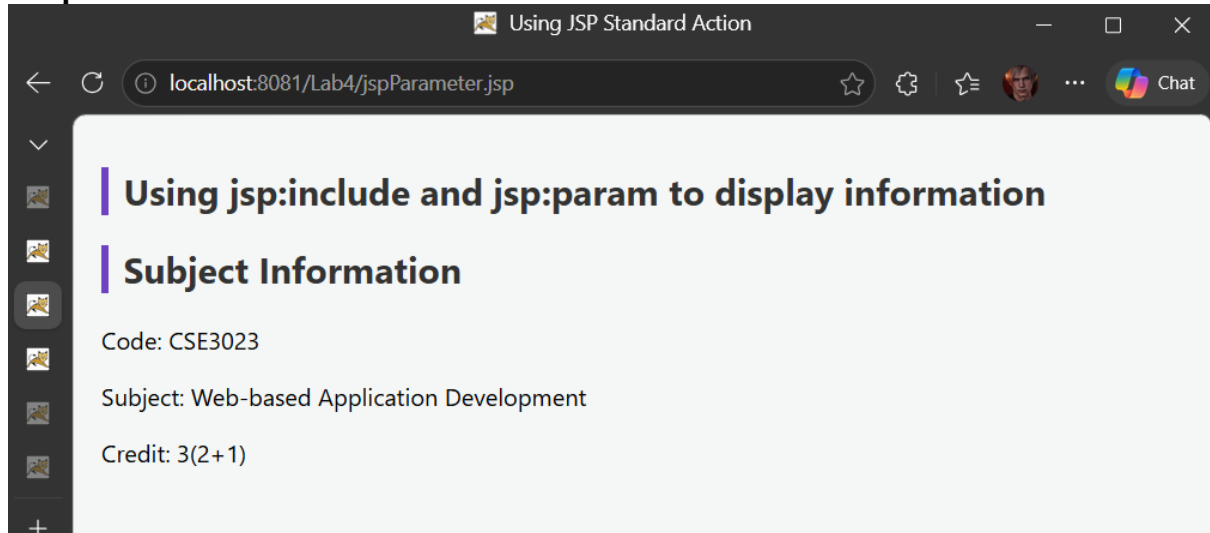
Objective: Apply and to reuse JSP components and pass data between pages

Answer:

```
...ml customer.css x processCustomer.jsp x jspParameter.jsp x currencyConversion.html x processCurrency.jsp x
Source History
<%--
2 Document : jspParameter
3 Created on : 21 Apr 2026, 3:44:00 pm
4 Author : syaheera
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11 <meta charset="UTF-8">
12 <title>Using JSP Standard Action</title>
13 <link rel="stylesheet" href="customer.css">
14 </head>
15 <body>
16 <h1>Using jsp:include and jsp:param to display information</h1>
17
18 <%
19 String sCode = "CSE3023";
20 String sSubject = "Web-based Application Development";
21 String sCredit = "3 (2+1)";
22 %>
23
24 <jsp:include page="subjectInfo.jsp" flush="true">
25 <jsp:param name="code" value="<%= sCode %%" />
26 <jsp:param name="subject" value="<%= sSubject %%" />
27 <jsp:param name="credit" value="<%= sCredit %%" />
28 </jsp:include>
29
30 </body>
31 </html>
```

```
...css processCustomer.jsp x jspParameter.jsp x subjectInfo.jsp x currencyConversion.html x processCurrency.jsp
Source History
<%--
2 Document : subjectInfo
3 Created on : 21 Apr 2026, 3:46:05 pm
4 Author : User
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12 <title>Subject Information</title>
13 <link rel="stylesheet" href="customer.css">
14 </head>
15 <body>
16 <h1>Subject Information</h1>
17
18 <p>Code: <%= request.getParameter("code") %"></p>
19 <p>Subject: <%= request.getParameter("subject") %"></p>
20 <p>Credit: <%= request.getParameter("credit") %"></p>
21
22 </body>
23 </html>
24
```

Output:



Reflection

1. What have you learnt from this exercise?

Answer:

I learned how to utilize JSP Standard Actions to create a modular web design. Specifically, you learned how to use `<jsp:include>` to reuse components and `<jsp:param>` to share data between different pages during a request.

2. List TWO (2) other JSP Standard Action tags.

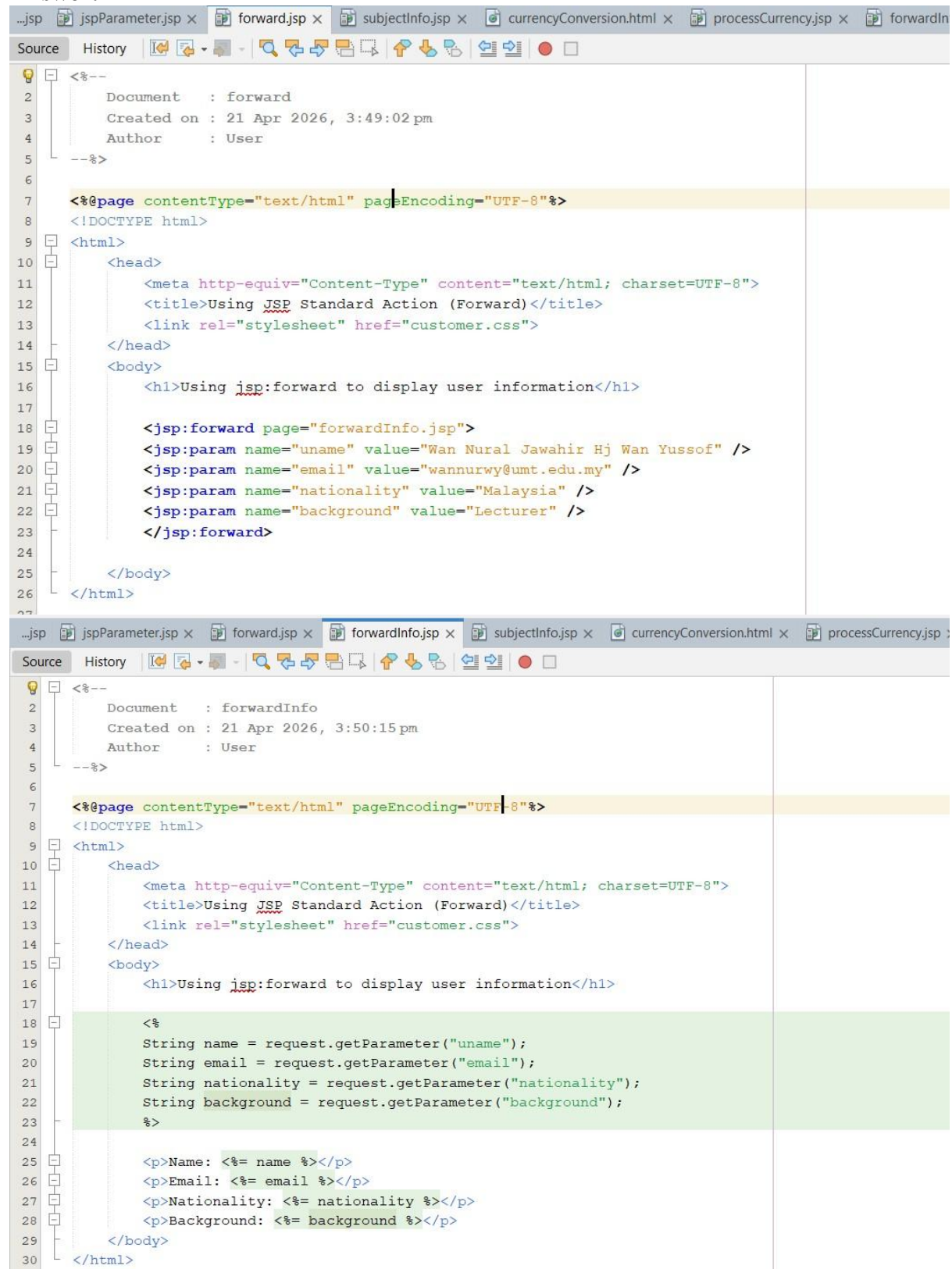
Answer:

- `<jsp:forward>`
- `<jsp:useBean>`

Task 4: Using JSP Standard Action (Forward)

Objective: Apply and to redirect requests and pass user data between JSP pages.

Answer:



```
...jsp  jspParameter.jsp x  forward.jsp x  subjectInfo.jsp x  currencyConversion.html x  processCurrency.jsp x  forwardIn

Source  History  [Icons]

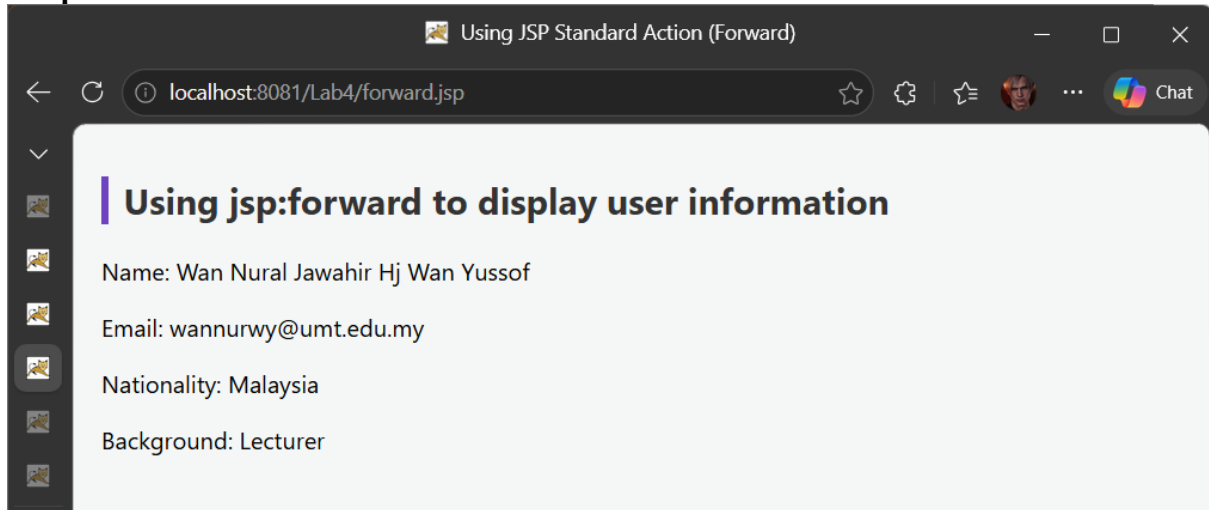
1  <%--
2      Document    : forward
3      Created on  : 21 Apr 2026, 3:49:02 pm
4      Author     : User
5  --%>
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <!DOCTYPE html>
9  <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Using JSP Standard Action (Forward)</title>
13         <link rel="stylesheet" href="customer.css">
14     </head>
15     <body>
16         <h1>Using jsp:forward to display user information</h1>
17
18         <jsp:forward page="forwardInfo.jsp">
19         <jsp:param name="uname" value="Wan Nural Jawahir Hj Wan Yussof" />
20         <jsp:param name="email" value="wannurwy@umt.edu.my" />
21         <jsp:param name="nationality" value="Malaysia" />
22         <jsp:param name="background" value="Lecturer" />
23     </jsp:forward>
24
25     </body>
26 </html>
27

...jsp  jspParameter.jsp x  forward.jsp x  forwardInfo.jsp x  subjectInfo.jsp x  currencyConversion.html x  processCurrency.jsp :

Source  History  [Icons]

1  <%--
2      Document    : forwardInfo
3      Created on  : 21 Apr 2026, 3:50:15 pm
4      Author     : User
5  --%>
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <!DOCTYPE html>
9  <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Using JSP Standard Action (Forward)</title>
13         <link rel="stylesheet" href="customer.css">
14     </head>
15     <body>
16         <h1>Using jsp:forward to display user information</h1>
17
18         <%
19             String name = request.getParameter("uname");
20             String email = request.getParameter("email");
21             String nationality = request.getParameter("nationality");
22             String background = request.getParameter("background");
23             %>
24
25         <p>Name: <%= name %></p>
26         <p>Email: <%= email %></p>
27         <p>Nationality: <%= nationality %></p>
28         <p>Background: <%= background %></p>
29     </body>
30 </html>
```

Output:



Task 5: Using JSP Scriptlet to Construct Business Logic

Objective: Apply JSP Scriptlet to implement business logic for an insurance quotation system, including calculation of insurance premium, NCD deduction, and SST charges

Answer:

```
...jsp  currencyConversion.html x  processCurrency.jsp x  processInsuranceQuo.jsp x  insuranceQuotation.jsp x  index.html x
Source  History  C:\Users\User\Documents\TAHUN 2 SEM 4\Lab4\web\insurance

<%--
2      Document    : insuranceQuotation
3      Created on  : 21 Apr 2026, 3:51:46 pm
4      Author     : syaheera
5  --%>
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <!DOCTYPE html>
9  <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Insurance Quotation</title>
13         <link rel="stylesheet" href="customer.css">
14     </head>
15     <body>
16         <h1>Insurance Quotation</h1>
17
18         <div class="card">
19             <h2 class="form-title">Customer Discount</h2>
20
21             <form action="processInsuranceQuo.jsp" method="post">
22
23                 <div class="form-group">
24                     <label>IC No</label>
25                     <input type="text" name="icno" placeholder="000000-00-0000" required>
26                 </div>
27
28                 <div class="form-group">
29                     <label>Name</label>
30                     <input type="text" name="name" placeholder="name" required>
31                 </div>
32
33                 <div class="form-group">
34                     <label>Market Price</label>
35                     <input type="number" name="price" step="0.01" placeholder="Price" required>
36                 </div>
37
38                 <div class="form-group">
39                     <label>Coverage Type</label>
40                     <select name="coverage">
41                         <option value="Comprehensive">Comprehensive</option>
42                         <option value="Third Party">Third Party</option>
43                     </select>
44                 </div>
45
46                 <div class="form-group">
47                     <label>No Claims Discount (NCD)</label>
48                     <select name="ncd">
49                         <option value="0.10">10%</option>
50                         <option value="0.25">25%</option>
51                         <option value="0.55">55%</option>
52                     </select>
53                 </div>
54
55                 <button type="submit">Submit</button>
56             </form>
57
58         </body>
59     </html>
60
```



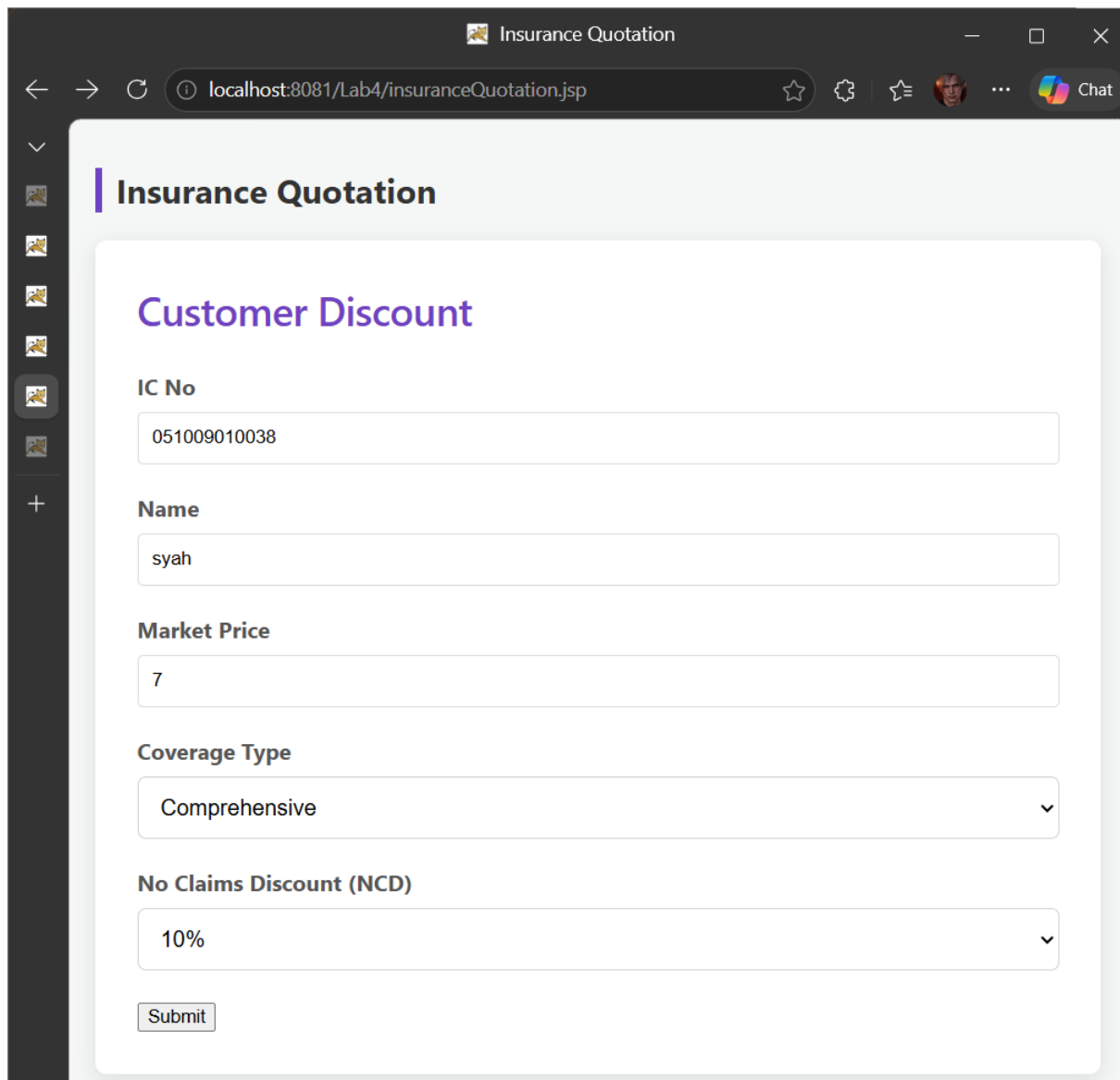
```
..jsp | currencyConversion.html x | processCurrency.jsp x | processInsuranceQuo.jsp x | insuranceQuotation.jsp x | index.htm
Source | History |
1 | <%--
2 |     Document    : processInsuranceQuo
3 |     Created on  : 21 Apr 2026, 4:20:52 pm
4 |     Author     : syaheera
5 | --%>
6 |
7 | <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 | <!DOCTYPE html>
9 | <html>
10 |     <head>
11 |         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12 |         <title>Process Insurance Quo</title>
13 |         <link rel="stylesheet" href="customer.css">
14 |     </head>
15 |     <body>
16 |         <h1>Process Insurance Quo</h1>
17 |
18 |         <%
19 |             // Retrieve form data
20 |             String icno = request.getParameter("icno");
21 |             String name = request.getParameter("name");
22 |             String coverage = request.getParameter("coverage");
23 |             String ncdStr = request.getParameter("ncd");
24 |             double price = 0;
25 |             double ncd = 0;
26 |             try {
27 |                 price = Double.parseDouble(request.getParameter("price"));
28 |                 ncd = Double.parseDouble(ncdStr);
29 |             } catch (Exception e) {
30 |                 price = 0;
31 |                 ncd = 0;
32 |             }
33 |             // Business Logic
34 |             double rate = 0;
35 |             String coverageDisplay = "";
36 |             if ("comprehensive".equals(coverage)) {
```

```

37     rate = 0.05; // 5%
38     coverageDisplay = "Comprehensive";
39   } else {
40     rate = 0.03; // 3%
41     coverageDisplay = "Third Party";
42   }
43   // Base insurance calculation
44   double insurance = price * rate;
45   // Apply NCD discount
46   double discount = insurance * ncd;
47   double afterNCD = insurance - discount;
48   // Add 8% SST
49   double sst = afterNCD * 0.08;
50   double finalAmount = afterNCD + sst;
51   %>
52
53   <div class="card" style="padding: 20px; border: 1px solid #ddd; border-radius: 8px; max-width: 400px;">
54
55     <p><strong>IC No:</strong><br>
56     <%= icno %></p>
57
58     <p><strong>Name:</strong><br>
59     <%= name %></p>
60
61     <p><strong>Market Price (RM):</strong><br>
62     <%= String.format("%.2f", price) %></p>
63
64     <p><strong>Coverage Type:</strong><br>
65     <%= coverage %></p>
66
67     <p><strong>NCD:</strong><br>
68     <%= (int)(ncd * 100) %>%</p>
69
70     <hr style="border: 0; border-top: 1px solid #000;">
71
72     <p><strong>Base Insurance:</strong><br>
73
74     <p><strong>NCD Discount:</strong><br>
75     RM <%= String.format("%.2f", discount) %></p>
76
77     <p><strong>After NCD:</strong><br>
78     RM <%= String.format("%.2f", afterNCD) %></p>
79
80     <p><strong>SST (8%):</strong><br>
81     RM <%= String.format("%.2f", sst) %></p>
82
83     <p><strong>Final Insurance Amount:</strong><br>
84     <span style="font-weight: bold; font-size: 1.1em;">RM <%= String.format("%.2f", finalAmount) %></span></p>
85
86     <a href="insuranceQuotation.jsp">
87       <button style="padding: 5px 20px; cursor: pointer;">Back</button>
88     </a>
89   </div>
90
91 </body>
92 </html>

```

Output:



The screenshot shows a web browser window with the title "Insurance Quotation". The address bar displays "localhost:8081/Lab4/insuranceQuotation.jsp". The page features a dark sidebar on the left with a vertical list of icons, including a plus sign at the bottom. The main content area has a header "Insurance Quotation" with a purple vertical bar to its left. Below the header is a section titled "Customer Discount" in purple. This section contains five form fields: "IC No" with the value "051009010038", "Name" with the value "syah", "Market Price" with the value "7", "Coverage Type" with a dropdown menu showing "Comprehensive", and "No Claims Discount (NCD)" with a dropdown menu showing "10%". Each dropdown menu has a small downward arrow icon on the right. At the bottom of the form is a "Submit" button.

Insurance Quotation

Customer Discount

IC No

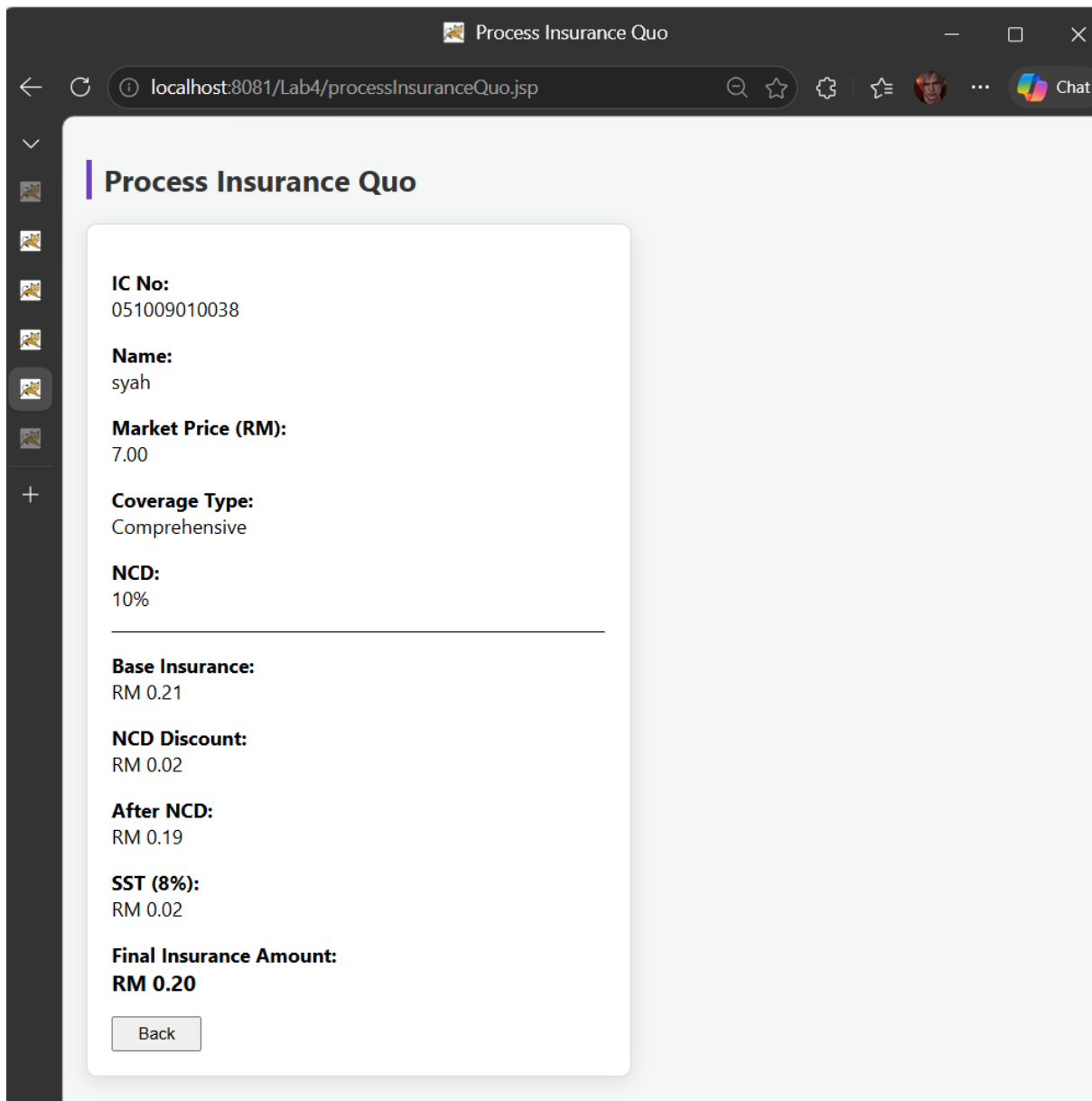
Name

Market Price

Coverage Type

No Claims Discount (NCD)

Submit



Reflection

1. What have you learnt from this exercise?

Answer:

This task taught how to develop a complete, interactive web application that integrates scripting elements with complex business logic, such as calculating premiums, applying deductions like NCD, and handling taxes like SST.

2. List TWO (2) more JSP Standard Action Tag

Answer:

- Variable Declarations and Assignments: Creating local variables to store data.
- Data Type Parsing: Using `Double.parseDouble()` or `Integer.parseInt()` to convert strings to numbers.
- Conditional Statements:

4.8 Exercise

Objective: Develop a BMI Calculator Web Application by integrating JSP Scriptlets, Expressions, Page Directives, Include Directives, and JSP Standard Actions (include, param, and forward) as learned in this lab.

Case Study

Faculty of Computer Science and Mathematics, UMT is organizing a Healthy Lifestyle Awareness Program. A web-based BMI Calculator System is required to help students calculate their Body Mass Index (BMI) and determine their health category instantly. As a web developer, you are required to develop a JSP-based application that demonstrates modular design, data passing, and dynamic processing using JSP technologies.

Instruction

Develop the system by fulfilling the following requirements:

1. Create a reusable header.jsp and footer.jsp.
2. Apply the JSP Include Directive () to include the header and footer in all JSP pages.
3. Design a navigation menu in the header that links to:
 - BMI Calculator Page (bmiCalculator.jsp)
 - Result Page • Health Information Page
4. Create a BMI input form in bmiCalculator.jsp:
 - Weight (kg)
 - Height (m)
5. Submit the form to processBMI.jsp and:
 - Retrieve values using request.getParameter()
 - Use JSP Scriptlet to:
 - Convert input values
 - Calculate BMI: $BMI = \frac{weight}{height^2}$
6. Use JSP Standard Action to forward the result to resultBMI.jsp.
7. Pass BMI value and category using . 132
8. In resultBMI.jsp:
 - Display BMI using JSP Expression
 - Format output to 2 decimal places
 - Display BMI category: – $BMI < 18.5$: Underweight – $18.5 \leq BMI \leq 25$: Normal – $BMI > 25$: Overweight
9. Create healthInfo.jsp:
 - Use JSP Page Directive to import Java classes
 - Store BMI categories using an ArrayList
 - Display the categories dynamically in a table
10. Ensure proper validation:
 - Input must be numeric
 - Height must not be zero

Answer:

```
...jsp  insuranceQuotation.jsp x index.html x header.jsp x footer.jsp x bmiCalculator.jsp x processBMI.jsp x res
Source History
<!--
2      Document    : footer
3      Created on  : 21 Apr 2026, 7:59:15 pm
4      Author     : syaheera
5  -->
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <!DOCTYPE html>
9  <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Footer</title>
13     </head>
14     <body>
15         <hr>
16         <div style="text-align: center; font-family: sans-serif; padding: 10px; color: #7f8c8d;">
17             <p>&copy; 2026 Faculty of Computer Science and Mathematics, UMT</p>
18         </div>
19     </body>
20 </html>
21
```

```
...jsp  insuranceQuotation.jsp x index.html x header.jsp x footer.jsp x bmiCalculator.jsp x processBMI.jsp x
Source History
<!--
2      Document    : header
3      Created on  : 21 Apr 2026, 7:58:24 pm
4      Author     : Nursyaheera binti mohd hisham
5  -->
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <!DOCTYPE html>
9  <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Header</title>
13     </head>
14     <body>
15         <div style="background-color: #2c3e50; color: white; padding: 20px; text-align:
16         center; font-family: sans-serif;">
17             <h1>Healthy Lifestyle Awareness Program</h1>
18             <nav>
19                 <a href="bmiCalculator.jsp" style="color: white; margin: 0 15px;
20                 text-decoration: none;">BMI Calculator</a>
21                 <a href="healthInfo.jsp" style="color: white; margin: 0 15px;
22                 text-decoration: none;">Health Information</a>
23             </nav>
24         </div>
25         <hr>
26     </body>
27 </html>
28
```

```
...jsp | insuranceQuotation.jsp x | index.html x | header.jsp x | footer.jsp x | bmiCalculator.jsp x | processBMI.jsp x | resultBMI.jsp...
Source | History | [Icons]
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11 <title>BMI Calculator</title>
12 <style>
13 .form-container { margin: 20px auto; width: 300px; font-family: sans-serif; }
14 input { width: 100%; padding: 8px; margin: 10px 0; }
15 </style>
16 </head>
17 <body>
18 <%%include file="header.jsp" %>
19
20 <div class="form-container">
21 <h2>BMI Input Form</h2>
22 <form action="processBMI.jsp" method="POST">
23 <label>Weight (kg):</label>
24 <input type="number" name="weight" step="0.1" placeholder="e.g. 70.5" required>
25
26 <label>Height (m):</label>
27 <input type="number" name="height" step="0.01" placeholder="e.g. 1.75" required>
28
29 <button type="submit" style="width: 100%; padding: 10px; background: #27ae60;
30 color: white; border: none; cursor: pointer;">
31 Calculate BMI
32 </button>
33 </form>
34 </div>
35
36 <%%include file="footer.jsp" %>
37 </body>
38 </html>
```

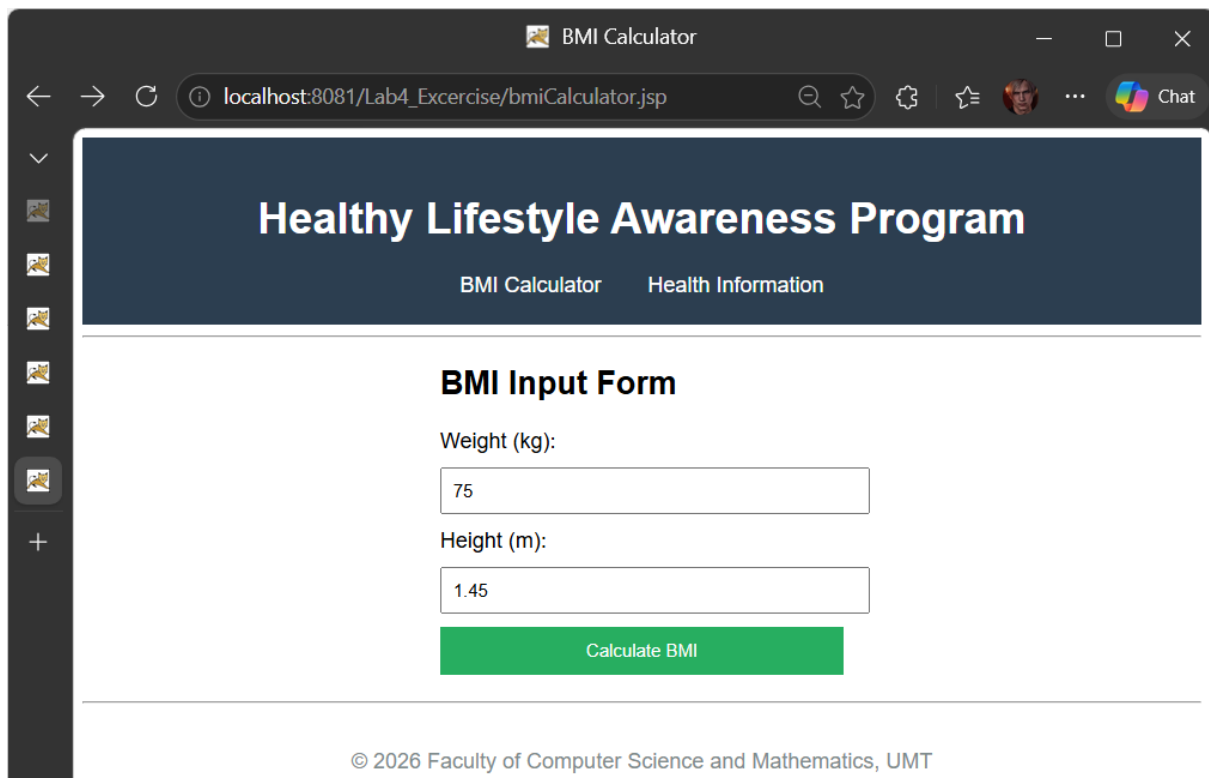
```
..jsp index.html x header.jsp x footer.jsp x bmiCalculator.jsp x processBMI.jsp x resultBMI.jsp x healthInfo.jsp x
Source History
1 <!--
2 Document : healthInfo
3 Created on : 21 Apr 2026, 8:00:49 pm
4 Author : sayheera
5 -->
6
7 <%@page import="java.util.ArrayList"%>
8 <%@page contentType="text/html" pageEncoding="UTF-8"%>
9 <!DOCTYPE html>
10 <html>
11 <head>
12 <title>Health Information</title>
13 <style>
14 table { width: 50%; margin: 20px auto; border-collapse: collapse; font-family: sans-serif; }
15 th, td { border: 1px solid #bdc3c7; padding: 12px; text-align: left; }
16 th { background-color: #ecf0f1; }
17 </style>
18 </head>
19 <body>
20 <%@include file="header.jsp" %>
21
22 <h2 style="text-align: center;">BMI Categories Reference</h2>
23
24 <%
25 // Dynamic ArrayList to store data
26 ArrayList<String[]> healthList = new ArrayList<>();
27 healthList.add(new String[]{"Underweight", "Below 18.5"});
28 healthList.add(new String[]{"Normal", "18.5 - 25.0"});
29 healthList.add(new String[]{"Overweight", "Above 25.0"});
30 %>
31
32 <table>
33 <tr>
34 <th>Category</th>
35 <th>Range</th>
36 </tr>
37
38 <tr>
39 <td><%= row[0] %></td>
40 <td><%= row[1] %></td>
41 </tr>
42 <% } %>
43 </table>
44
45 <%@include file="footer.jsp" %>
46 </body>
47 </html>
48
```



```
...jsp index.html x header.jsp x footer.jsp x bmiCalculator.jsp x processBMI.jsp x resultBMI.jsp x healthInfo.jsp >
Source History
1 <!--
2 Document : processBMI
3 Created on : 21 Apr 2026, 8:00:09 pm
4 Author : sayheera
5 -->
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <%
9 // Retrieve values
10 String weightStr = request.getParameter("weight");
11 String heightStr = request.getParameter("height");
12
13 try {
14     double weight = Double.parseDouble(weightStr);
15     double height = Double.parseDouble(heightStr);
16
17     // Validation: Height must not be zero
18     if (height <= 0) {
19         double bmi = weight / (height * height);
20         String cat = (bmi < 18.5) ? "Underweight" : "Normal";
21     } else {
22         // Calculate BMI: weight / (height * height)
23         double bmi = weight / (height * height);
24
25         // Determine Category
26         String category;
27         if (bmi < 18.5) {
28             category = "Underweight";
29         } else if (bmi <= 25) {
30             category = "Normal";
31         } else {
32             category = "Overweight";
33         }
34     }
35
36     <!-- Forward to result page with parameters -->
37     <jsp:forward page="resultBMI.jsp">
38         <jsp:param name="bmiValue" value="<%= String.valueOf(bmi) %>" />
39         <jsp:param name="category" value="<%= category %>" />
40     </jsp:forward>
41
42 } catch (NumberFormatException e) {
43     out.println("<script>alert('Please enter valid numeric values!'); history.back();</script>");
44 }
45 %>
```

```
...jsp index.html x header.jsp x footer.jsp x bmiCalculator.jsp x processBMI.jsp x resultBMI.jsp x health
Source History
1 <%--
2     Document    : resultBMI
3     Created on  : 21 Apr 2026, 8:00:29 pm
4     Author     : syaheera
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11     <title>BMI Result</title>
12     <style>
13         .result-box { text-align: center; margin-top: 30px; font-family: sans-serif; }
14         .bmi-val { font-size: 2em; color: #2980b9; font-weight: bold; }
15     </style>
16 </head>
17 <body>
18     <%@include file="header.jsp" %>
19
20     <div class="result-box">
21         <h2>Your BMI Result</h2>
22
23         <%
24             String bmiStr = request.getParameter("bmiValue");
25             String category = request.getParameter("category");
26
27             // Default to 0 if null to avoid errors
28             double bmi = (bmiStr != null) ? Double.parseDouble(bmiStr) : 0.0;
29             if (category == null) category = "Unknown";
30         %>
31
32         <p>Your calculated BMI is:</p>
33         <div class="bmi-val">
34             <%= String.format("%.2f", bmi) %>
35         </div>
36
37         <br>
38
39         <a href="bmiCalculator.jsp">Back to Calculator</a>
40     </div>
41
42     <%@include file="footer.jsp" %>
43 </body>
44 </html>
45
46
```

Output:



The screenshot shows a web browser window titled "BMI Calculator". The address bar displays "localhost:8081/Lab4_Excercise/bmiCalculator.jsp". The page features a dark blue header with the title "Healthy Lifestyle Awareness Program" and two navigation links: "BMI Calculator" and "Health Information". Below the header, the main content area is titled "BMI Input Form". It contains two input fields: "Weight (kg)" with the value "75" and "Height (m)" with the value "1.45". A green button labeled "Calculate BMI" is positioned below the input fields. At the bottom of the page, a footer displays the copyright notice "© 2026 Faculty of Computer Science and Mathematics, UMT".

BMI Calculator

Healthy Lifestyle Awareness Program

BMI Calculator Health Information

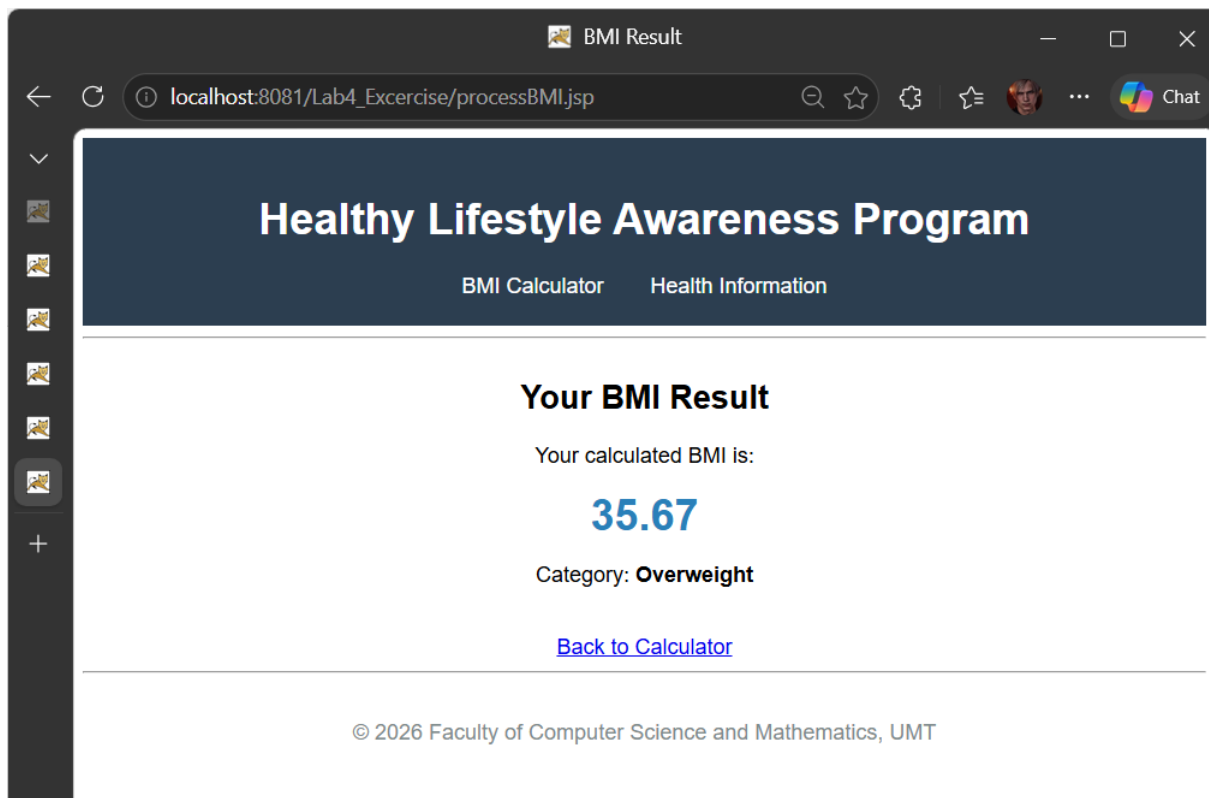
BMI Input Form

Weight (kg):

Height (m):

Calculate BMI

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The screenshot shows a web browser window titled "BMI Result". The address bar displays "localhost:8081/Lab4_Excercise/processBMI.jsp". The page features a dark blue header with the title "Healthy Lifestyle Awareness Program" and two navigation links: "BMI Calculator" and "Health Information". Below the header, the main content area is titled "Your BMI Result". It displays the text "Your calculated BMI is:" followed by the value "35.67" in large blue font. Below this, the text "Category: Overweight" is shown. A blue link labeled "Back to Calculator" is positioned at the bottom of the main content area. At the bottom of the page, a footer displays the copyright notice "© 2026 Faculty of Computer Science and Mathematics, UMT".

BMI Result

Healthy Lifestyle Awareness Program

BMI Calculator Health Information

Your BMI Result

Your calculated BMI is:

35.67

Category: **Overweight**

[Back to Calculator](#)

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